INCH-POUND

MIL-C-43063F
14 January 1991
SUPERSEDING
MIL-C-43063E
30 June 1986

MILITARY SPECIFICATION

COVERALLS, ROCKET FUEL HANDLER'S, IMPERMEABLE, FULL PROTECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one-piece, impermeable, rocket fuel handler's coveralls for full protection from fuels and oxidizers. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The coveralls shall be of one type in the following sizes as specified (see 6.2).

Small	Medium	Large	X-Large	XX-Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-169	- Calcium Chloride, Technical
A-A-203	- Paper, Kraft, Untreated
T-C-571	- Cords, Cotton; General and Special Purposes, Sash and Venetian Blind
V-F-106	- Fastener, Slide, Interlocking
V-T-276	- Thread, Cotton
JJ-W-155	- Webbing, Textile, (Cotton, Elastic)
UU-P-328	- Paper, Manifold
DDD-L-20	- Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, and JP-5/JP-8 ST
MIL-P-7254	- Propellant, Nitric Acid
MIL-F-10884	- Fasteners, Snap
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistance Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43062	- Cloth, Coated, Nylon, Resin Modified Butyl-Coated, Acid and Fuel Resistant
MIL-P-51493	- Paper, Chemical Agents Detector: M9

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams and Stitchings

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-1-3 - Jig, Acid Testing, No. 2
8-1-6 - Coveralls, Rocket Fuel Handlers; Closure Ring, Sleeve
2-1-293 - Coveralls, Rocket Fuel Handler's, Impermeable, Full Protection; Method of Folding and Positioning

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case the specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be unvulcanized, resin modified butyl coated nylon conforming to MIL-C-43062.

3.3.2 Strapping and flap reinforcement. The gum strapping used to seal the seams and for reinforcing the front protective flap shall be made from an unsupported film of the uncured coating compound used to coat the fabric. The gum strapping shall be 0.025 $\pm$ 0.005 inch thick and thoroughly clean before application to any seam. The gum strapping for seams shall be 3/4 $\pm$ 1/8 inch wide; the strapping for sleeve wristlet shall be 1-1/4 $\pm$ 1/8 inches wide; and the flap reinforcement shall be cut to pattern dimensions (see 4.4.1.1).

3.3.3 Cement. The cement used to seal the seams and the strappings over the seams shall be a curing modified butyl base cement. If the finished seam conforms to the requirements of 3.3.12 and table II, (see 3.3.13) the cement shall be considered satisfactory. The color of the cement, when dry, shall approximate that of the coated cloth, otherwise a clear cement shall be used.

3.3.4 Webbing, elastic. The elastic webbing shall be cotton elastic, conforming to type I, class 3, 3/4 inch wide, of JJ-W-155, or a woven or braided elastic webbing of equivalent grade may be used, except that the colorfastness requirements shall not apply. The elastic webbing may be furnished in black or Olive Drab 7.

3.3.5 Thread. The thread for seaming and stitching the coveralls shall be cotton type IA3, ticket No. 30, 4 ply conforming to V-T-276. The thread shall be dyed black. Colorfastness requirements shall not apply.

3.3.6 Cotton cord. The cord for reinforcing neck facing shall be cotton, conforming to type I, class 1, size 4 of T-C-571.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

3.3.7 Label. Each of the coveralls shall have a label containing the following information in sequence: size, identification, and instructions. The label shall be applied to the inside upperback portion of the coveralls in accordance with type IV, class 14 of DDD-L-20. The marking medium shall be white, yellow, or light gray. The label may be marked on a piece of the uncured fabric specified in 3.3.1, then cemented on the coveralls and cured with the coveralls after assembly. The printed label shall be well defined and clearly legible after being tested for crocking, both wet and dry, when tested as specified in 4.4.5.

3.3.7.1 Instruction legend. The instructions shall be as follows:

1. Wear for protection against injury from dangerous liquid fuels and oxidizers. DO NOT USE for firefighting.

2. DO NOT USE the same coverall for fuels and oxidizers without washing between usage.

3. Put on over the clothing required to keep warm. DO NOT wear over bare skin.

4. Wear size that fits loosely with no binding. Coveralls are oversized to fit over cold weather clothing.

5. Keep slide fastener and snap closure closed to prevent entry of harmful agents.

6. Roll back rubber wristlet, insert sleeve ring tightly in glove ring, and roll down rubber sleeve wristlet to seal the joint. When coverall is not in use, wristlet MUST BE left unrolled.

7. Wear coverall legs over footgear. DO NOT tuck coverall cuffs inside boot.

8. To prevent overheating, put on the cooling coverall, and wet down frequently.

9. CAUTION: If dangerous agents are spilled on coverall, wash down immediately. Use large amounts of water for a thorough rinse.

10. Washing shall be accomplished in accordance with TM 10-277.

3.3.8 Sleeve wristlet. The finished sleeve wristlet shall be cylindrical in shape and shall have a beaded or rolled edge 1/4 to 3/8 inch in diameter on one end. The wristlet shall be fabricated from unsupported film of the coating compound used to coat the coated fabric. The finished wristlet shall be 0.025 $\pm$ 0.005 inch thick, 3-3/4 $\pm$ 1/8 inches in diameter, and shall be 3 $\pm$ 1/8 inches in length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

3.3.9 Sleeve closure ring. The sleeve closure ring shall be molded from virgin, unfilled, natural color polypropylene conforming to the requirements in table I. The molded pieces shall be uniform in color, texture, and finish; and shall be free from cracks, blisters, scratches and other defects affecting serviceability. The dimensions of the molded polypropylene ring shall conform to Drawing 8-1-6 (4.4.1.1).

TABLE I. Polypropylene material requirements

Characteristic	Requirement	
	Minimum	Maximum
Specific gravity	0.895	0.904
Tensile strength (psi)	4,000	---
Distortion under heat at 66 p.s.i: (Fiberstress) °F	190	---

3.3.10 Snap fasteners. Snap fasteners shall consist of a size 1 or 2, 24 line size button with socket for the female half and a size 1 or 2 eyelet with stud for the male half conforming to style 2, finish 2 of MIL-F-10884.

3.3.11 Slide fastener. The slide fastener shall be type I, style 3, size MHS, slider with wire stirrup pull, 36 inches long for all sizes and shall conform to V-F-106. The fastener chain shall be fabricated of brass. The tape shall be dyed black and shall be water repellent treated. Colorfastness requirements shall not apply.

3.3.12 Sealed seams. All seams on the outer surface of the coveralls shall be sealed with a strapping as specified in 3.3.2. After the seam has been cemented, the strapping shall be centered and cemented over the raw edge of the seam on the outside. The cement line may extend a maximum of 3/4 inch on either side of the strapping. After the coveralls have been completely assembled, it shall be fully vulcanized.

3.3.12.1 Acid resistance. The completed seams and surrounding area shall show no penetration as indicated by change of color of the detector paper, no evidence of cracking, stiffening, flaking, swelling, separation of the strapping from seam, separation of the coating from the fabric, or surface tack rating greater than II, when tested as specified in 4.4.5.

3.3.12.2 Fuel resistance. The completed seams and surrounding area shall show no evidence of cracking, stiffening, flaking, swelling, separation of the strapping from the seam, separation of the coating from the fabric, or surface tack rating greater than II, when tested as specified in 4.4.5.

3.3.13 Seam and seam strapping adhesion. Adhesion of the seam and seam strapping for all seams shall conform to requirements of table II, when tested as specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE II. Seam and seam strapping requirements

	Requirement (minimum)
Seam gum strapping adhesion	3.5 lbs/inch
Seam gum strapping adhesion after immersion	3.0 lbs/inch
Seam gum strapping adhesion after heat aging	3.0 lbs/inch
Cemented seam adhesion	3.5 lbs/inch
Cemented seam adhesion after immersion	3.0 lbs/inch
Cemented seam adhesion after heat aging	3.0 lbs/inch
Resistance to cold crack of finished cemented and strapped seam	1/
Seam hydrostatic resistance of finished cemented and strapped seam	70 p.s.i.

1/ The seam shall show no signs of cracking, flaking, separation, stiffening, delamination or tackiness after this test has been performed.

3.4 Design. The coveralls (see figure 1) shall have a slide fastener closure with a pullman type snap closure over the fastener to ensure complete protection. Each sleeve shall have a closure ring insert and a rolled sleeve wristlet which rolls down over the ring of the glove after glove has been properly attached.

3.5 Patterns. Standard patterns, as listed below, which provide a 5/8 inch allowance for all seams will be furnished by the Government to the contractor for cutting working patterns. The Government patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's work patterns. The working patterns shall be identical to the Government patterns.

List of pattern parts

Material	Nomenclature of parts	Cut parts
Cloth, modified butyl coated	Right front	1
	Left front	1
	Back	2
	Sleeve	2
	Right front facing	1
	Left front facing	1
	Right front flap	1
	Left front flap	1
	Right neck facing	1
	Left neck facing	1
	Leg strap cover	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

List of pattern parts (cont'd)

Material	Nomenclature of parts	Cut parts
Cloth, modified butyl coated (cont'd)	Sleeve facing	2
	Leg reinforcement	2
	Bottom of slide fastener stay reinforcement	1
	Bottom fly stay	1
	Leg adjustment strap reinforcement	2
Gum strapping compound	Left front protective flap reinforcement	1

3.6 Construction. The construction shall conform in all respects to the requirements specified in table III and herein. Figure 1 is furnished solely for guidance and information. Should variation from the written specification appear in figure 1, the specification shall govern.

3.6.1 Stitches, seams and stitching. Stitch, seams, and stitching types as specified in table III shall conform to FED-STD-751.

3.6.1.1 Seam types. Seam types shall be as specified in table III, with seam allowances maintained and seams properly cemented and strapped so that no raw edges, twists, pleats, puckers, or open seams will result.

3.6.1.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table III.

3.6.1.3 Type 301 stitching. Ends of all stitchings shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose bobbin or top thread, or excessive tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.6.1.3.1 Repairs of 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, runoffs, or bobbin runouts occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.i.4 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.6.2 Assembly. The coveralls shall be completely assembled, using an unvulcanized coated fabric (3.3.1), with all outside seams strapped, using an unvulcanized strapping as specified in 3.3.2. The cement for adherence of seams and strapping shall be as specified in 3.3.3. After assembly, the coveralls shall be dusted with an inert dusting powder then fully vulcanized. The assembled and vulcanized coveralls may be cleaned with rubber makers naptha or other suitable cleaning agent which is not a solvent for the cured coating compound.

3.6.3 Repairs in coating. Prior to vulcanization, areas containing coating defects may be repaired by application of a round patch 1-1/2 $\pm$ 1/4 inches in diameter using the strapping and flap reinforcement gum strapping specified in 3.3.2 and the cement specified in 3.3.3 applied and cured as specified for the sealed seams in 3.3.12. The patch shall be cemented to the coveralls and the patch shall extend a minimum of 1/2 inch beyond all edges of the damaged area, and shall be well adhered to the coated fabric so that when subjected to flexing action between the hands, will show no lifting of the edge, and no cracking, flaking or removal of the cement or patch. The repaired area shall show no stiffening or tackiness which would affect serviceability. Repairing of fabric coating defects by methods other than patching as indicated above are not authorized.

3.6.3.1 Location and number of repairs. Patches not exceeding six in number may be made to each of the coveralls, provided not more than two patches are made on any of the six major exterior pattern parts of the coveralls which are the left front, back and sleeve, right front, back and sleeve.

3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with operation requirements specified in table III. The contractor is not required to follow the exact sequence of operations, except for operations 19 (when label is stamped) 20 and 21, which shall be performed after the suit has been vulcanized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

NO.	TABLE III. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the coveralls in accordance with patterns furnished, which show size, shape, and notches for proper assembly. A tolerance of 1-1/2 inches maximum from the directional line on the pattern is acceptable. b. Cut the butyl coated cloth so that the heavier coated side of the cloth will form the outer surface of the coveralls. c. Cut the body panels from one piece of material; the sleeves and reinforcement pieces may be cut from ends. d. Cut elastic webbing 6-1/2 ± 1/4 inches long.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Make fronts. Join right and left front parts from neck to bottom of crotch by overlapping the front edges 5/8 ± 1/8 inch securely cementing the seam overlap, and strap on the outside the length of seam with gum strapping, centered on raw edge of seam.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Slide fastener opening. a. Cut slide fastener opening, cutting right and left front, as indicated by marks on pattern. b. Cement outside and inside area along left and right sides of cut opening to receive slide fastener, facing and protective flaps. c. Thoroughly cement slide fastener tape surface with not less than three coats of cement (or until weave of tape is indistinguishable). Metal scoops shall be free of cement.						
5.	Attach slide fastener and inside facings to front opening. a. Attach right and left side of slide fastener to right and left front respectively on inside of cut opening. Edge of cut opening shall measure 3/16 ± 1/16 inch from metal scoops of slide fastener. Bottom of slide fastener shall be positioned at bottom of cut opening for all sizes. Turn tape extension of slide fastener at top edges, if required for size Small. b. Securely cement right and left facings to inside of coveralls with edge of facings 3/16 ± 1/16 inch from metal scoops of slide fastener and top edge of facing even with top edge of coveralls.						

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make and attach protector flaps.						
	a. Securely cement right protector flap.						
	b. Hem top edge of right protector flap 1/4 ± 1/8 inch by folding to the inside. Securely cement hemmed portion.						
	c. Place a suitable separator (5/8 ± 1/8 inch wide) at raw edge the full length and fold flap in half lengthwise securely cementing both plies together, except the separated portion.						
	d. Remove separator and spread open uncemented portion of flap and securely cement (one ply) to right front of coveralls with the edge of flap coinciding with cut opening and top of flap even with neck opening. Securely cement the other ply of flap to body of coveralls turning back the flap, forming a hinge.						
	e. Repeat operations a, b, and c above for left protector flap.						
	f. Securely cement protective flap reinforcement (of gum strapping compound) on outside of left flap as indicated by pattern, not more than 1/8 inch from folded edge.						
	g. Spread open uncemented portion of flap and securely cement (one ply) to left front of coveralls with the edge of flap coinciding with cut opening (± 1/8 inch) and top edges of flap coinciding with neck opening. Securely cement the other ply of flap to body of coveralls turning the flap forward, forming a hinge.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make and attach protector flaps. (cont'd)						
	h. Cemented area attaching protector flaps to coveralls shall measure 1-1/4 ± 1/4 inches wide.						
	i. Gum strap the outside edges of right and left protector flaps the entire length with the gum strapping centered over raw edge of seam.						
7.	Reinforce base of front opening.						
	a. Fold under long edge of left protector flap not less than 1 inch, inserting right protector flap in fold at bottom. Securely cement the reinforcement piece to base of opening with the one end extending over the left front protector flap. The reinforcement piece shall finish not less than 1-1/2 inches wide and centered (± 1/4 inch) on the base of the opening.						
	b. Securely cement reinforcement stay at center bottom of slide fastener on inside of coveralls.						
8.	Join side seams.						
	Lap the fronts 5/8 ± 1/8 inch over the backs and securely cement the overlapped seam and strap on the outside with gum strapping centered on the raw edge of seam.						

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Attach leg reinforcement. Position the leg reinforcement 1/2 ± 1/8 inch from bottom of each leg on inside, and securely cement the lighter coated side of the leg reinforcement to coveralls.						
10.	Make back. Lap the back parts on each other 5/8 ± 1/8 inch, securely cementing the overlapped seam, and securely strap the outside with gum strapping centered on raw edge of seam.						
11.	Join inseams. Join inseams from bottom of leg through crotch by overlapping the edges 5/8 ± 1/8 inch, securely cementing the overlapped seam and securely strap the outside with gum strapping centered on raw edge of seam.						
12.	Make sleeves. a. Join inseams of sleeves by overlapping the seams 5/8 ± 1/8 inch, securely cementing the overlapped seam and gum strap the outside with gum strapping centered on the raw edge of seam. b. Position sleeve facing on inside of sleeves and securely cement.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Set sleeves. a. Set sleeves to front and back parts by overlapping the seam 5/8 ± 1/8 inch and securely cementing the overlapped seams to the coverall parts. Inseam of sleeve and side seam of coveralls shall not be staggered more than 3/4 inch. NOTE: Overlap of seam may be reduced to not less than 3/8 inch at armpits to facilitate sleeve setting. b. Securely strap the overlapped seams with gum strapping on the outside with the gum strapping centered on raw edge of seam.						
14.	Set neck facing. a. Hem neck of coveralls 3/8 ± 1/8 inch (over cord) by folding raw edge to inside and securely cement; securely cementing the cotton cord reinforcement, cut to proper length, in the fold of the facing. b. Securely cement neck facings to neck of coveralls with the front edges of the fastener facings and with the top raw edge of facings positioned along base of cord ± 1/16 inch. The other end of the facings shall be trimmed so overlap in the back is no more than 1 inch.						

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Hem leg bottom. Turn under leg bottom 1/2 ± 1/8 inch and securely cement. Turn under shall be over leg reinforcements.						
16.	Make leg strap covers. a. Fold leg strap covers on itself in thirds, heavy coated side out, and securely cement the top ply to the middle ply. Inside of strap shall be thoroughly dusted to prevent adhesion during vulcanization. b. Make a 1 ± 1/8 inch vertical slit in the leg reinforcement patches. Pull leg straps through slits and securely cement patches to leg.						
17.	Attach snap fasteners. a. Attach seven stud parts of the snap fastener to the inside of protector flap of left front as indicated by marks on patterns. b. Attach seven socket parts of the snap fastener to right front, through strapping of the right front flap. The position of the socket parts shall correspond to stud parts on the left protector flap, resulting in a smooth flat closure. c. Attach two stud parts of the fastener through leg, and leg reinforcements as indicated by marks on pattern, ± 1/4 inch off tolerance from pattern marks permitted. d. Securely cement a patch of basic material, approximately 2-3/4 by 3/4 inches on inside of leg covering stud parts of snap fastener.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Join sleeve wristlet and sleeve closure rings to sleeve.						
	a. Insert sleeve closure ring (beaded end) into end of sleeve 7/8 ± 1/8 inch and securely cement ring to sleeve.						
	b. Stitch sleeve to closure ring 5/16 to 5/8 inch from top inner edge of ring.	301	SSa-1	6-8	30-4	30-4	
	c. Insert closure ring into sleeve wristlet so that the end of the closure ring is 9/16 ± 1/8 inch beyond bead of wristlet. Cement entire area of overlap from bottom edge of sleeve and raw edge of wristlet.						
	d. Cement and strap outside of wristlet raw edges with 1-1/4 ± 1/8 inches wide strapping, overlapping ends 1/2 to 5/8 inch at inseam of sleeve.						
19.	Attach label.						
	a. Stamp or cement label (centered) on inside right back not less than 3 inches down from top edge of neck.						
20.	Insert elastic into leg strap cover.						
	a. Insert elastic webbing into leg strap cover, so that end of elastic is under reinforcement patch.						
	b. Stitch strap through reinforcement patch and leg the width of strap with the stitching securely back-stitched.	301	SSa-1	8-12	30-4	30-4	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
20.	Insert elastic into leg strap cover. (cont'd)						
	c. Turn in raw edges of leg strap cover to inside 1/8 to 3/8 inch and secure with box stitching, catching elastic in turn-in. Stitching shall be 1/16 to 3/16 inch from edges.	301	SSa-1	8-12	30-4	30-4	
	d. Attach one socket part of the snap fastener in the center of the box stitching at the free end of each leg strap. The cap shall be on the top side when fastener is engaged.						
21.	Check coveralls.						
	a. Close and open slide fastener, repeating process at least three times.						
	b. Close all snap fasteners checking flap positioning, leg strap elongation, and uniformity of stud and socket positioning.						
	c. Release snap fasteners and check security of attachment.						
	d. Check all outside seams for security of strapping and seam coverage.						
	e. Thoroughly impregnate all thread exposed on outside of coveralls with a solution of the coating compound.						

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

3.8 Finished measurements. The finished measurements for the coveralls shall conform to table IV.

TABLE IV. Finished measurements (inches)

	Small	Medium	Large	X-Large	XX-Large	Tolerance
1/2 Chest 1/	21-3/4	23-1/2	25-1/4	27	28-3/4	± 3/4
Sleeve inseam 2/	22-1/4	23-1/2	24-3/4	26	27-1/4	± 3/4
Leg inseam 3/	28-1/2	28-1/2	28-1/2	28-1/2	28-1/2	± 1/2

1/ With the slide fastener closed, and front of coveralls smoothed flat, measure across front from a point 1/2 inch below armhole seam from side seam to side seam.

2/ Measure the sleeve inseam from the base of armhole along the inseam of sleeve to bottom of cuff.

3/ Measure from the base of crotch along the inside of leg to the bottom of leg.

3.9 Workmanship. The finished coveralls shall conform to the product quality established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. Utmost care shall be taken during component and subassembly fabrication to ensure quality workmanship and safety of the service person using the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. The contractor shall furnish a certificate of compliance stating that the sleeve closure ring material conforms to the requirements specified in 3.3.9, and the gum strapping and strapping for the sleeve wristlet conform to the requirements specified in 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 End item critical defect examination. Every end item, prior to the visual and dimensional examinations in 4.4.3 and 4.4.4, shall be examined for the critical defects listed in 4.4.3. Any end item found to contain one or more critical defects shall be rejected.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table V. The lot size shall be expressed in units of coveralls. The sample unit shall be one of the coveralls. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot.

TABLE V. End item visual defects

Defects	Classification		
	Critical	Major	Minor
Finish (applicable to all components):			
Not as specified			201
Cleanness:			
Cement line extends more than 3/4 inch on either side of the strapping			202
Spot, stain, or foreign matter, which cannot be readily removed with rubber markers naptha or other suitable cleaning agent			203
Material (general):			
Any component not fabricated of the specified material		101	
Coated fabric:			
Hole, cut, abraded area exposing base fabric, tear or rip through material	1		
Area not coated	2		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Coated fabric: (cont'd)			
Blister, tunnel, or delamination of coating	3		
Brittle area which when bent, results in a crack	4		
NOTE: If any of the above four conditions occur on a reinforcement patch, hem, or adjustment strap, it shall be classified as a minor defect.			
Area of excessive coating, i.e., solid runs, ridges or lumps			204
Patches or repairs in coated cloth			
Patch less than 1-1/4 inches		102	
Patch greater than 1-3/4 inches but not greater than 2 inches			205
Patch greater than 2 inches		103	
More than two patches per major pattern part			206
More than six patches per coverall		104	
Edge of patch lifting, cracking, flaking more than 1/16 inch or removal of the cement or patch when hand flexed as specified in 3.6.3	5		
Patch not smoothly and evenly adhered		105	
Patch area excessively stiff or tacky	6		
Patch extends less than 1/2 inch beyond all edges of the damaged area		106	
Repair made by method other than patching	7		
Pit or abraded area but not through material or to a degree where base fabric is exposed			207
Tacky, i.e., improper vulcanization		107	
Imbedded foreign material which when removed will expose base fabric or leave a through hole	8		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Coated fabric: (cont'd)			
Foreign inclusion which cannot be readily removed or if removable, does not expose base fabric			208
Color of cement not approximating color of coated cloth (not applicable to clear cement)			209
NOTE: A pick glass may be used to determine the degree of the deficiency.			
Cutting:			
Any part not cut in accordance with the directional lines indicated on pattern or in accordance with specification requirement		108	
Directional line on front or back part vary from warp direction by more than 1-1/2 inches		109	
Construction and workmanship (general):			
Component damaged, misplaced, operation omitted or not properly performed which may result in injury to the user	9		
Component damaged, misplaced, operation omitted or not properly performed seriously affecting serviceability		110	
Component damaged, misplaced, operation omitted or not properly performed affecting serviceability but not seriously			210
Heavy coated side not on exterior of coverall		111	
Stitching:			
Not type specified			211
Omitted at free end of elastic strap			212

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Stitching: (cont'd)			
Omitted at attached end of elastic strap		112	
Not securely stitched where required			213
Thread end not trimmed to 1/4 inch maximum length			214
Thread not black			215
Thread not impregnated with coating compound		113	
Assembly of vulcanized and cemented parts:			
Not joined in accordance with the specified construction	10		
Open seam completely through coverall	11		
Area of delamination other than on strapping, but does not constitute an opening completely through coverall			216
Area of delamination at edge of strapping 1/16 inch or more, but does not extend to the raw edge of seam			217
Area of delamination of strapping extending to the raw edge of the seam	12		
Any blister in the strapping		114	
Seam strapping off center on outer edge of seam lap by more than 1/4 inch			218
Seam strapping not covering raw edge of seam	13		
Leg reinforcements omitted	14		
Lighter coated side of leg reinforcement finished on inside of coverall		115	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Assembly of vulcanized and cemented parts: (cont'd)			
Any hole, cut, tear, or rip in the seam strapping, exposing base fabric	15		
NOTE: Extent of delamination (or tunnel) may be determined by insertion of any blunt end instrument suitable for this purpose. Penetration of the instrument shall cease immediately if resistance is met.			
Front opening:			
Slide fastener opening reversed, i.e., opening extends into right leg instead of left leg		116	
Protector flaps not two ply construction		117	
Protector flaps puckered or distorted but not to a degree where an effective closure cannot be accomplished			219
Facing omitted		118	
Reinforcement piece at base of front opening either on inside or outside omitted		119	
Left flap reinforcement omitted	16		
Hardware:			
Slide fastener:			
Not securely attached or affixed	17		
Puckered, distorted or twisted		120	
Will not operate or requires abnormal force to operate		121	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Hardware: (cont'd)			
Snap fastener:			
Any snap fastener missing or will not function on front opening		122	
Alignment of male portion of snap fastener and corresponding female portion of snap fastener more than 1/4 inch apart on front opening		123	
One stud part of fastener omitted on trouser leg			220
Both stud parts of fastener omitted on trouser leg		124	
Socket part of fastener omitted on leg strap		125	
Any sharp burrs or ridges		126	
Sleeves:			
Closure ring not inserted into end of sleeve			221
Facings omitted			222
Closure ring:			
Omitted		127	
Not as specified		128	
Sharp edges not removed from ends.		129	
Distorted, twisted or damaged		130	
Any blister, crack, fracture or other defect		131	
Not attached as specified		132	
Sleeve wristlet:			
Not beaded at the end		133	
Wristlet does not fit to plastic ring		134	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Defects	Classification Critical	Major	Minor
Label:			
Missing, incorrect, illegible, not in specified location, characters not of specified height, incomplete, not specified color, or not accomplished as required			223

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table IV. Any dimension not within the specified tolerance shall be classified as a defect. Inseams uneven in length by more than 1/2 inch shall be scored as a defect. The lot size shall be expressed in units of coveralls. The sample unit shall be one of the coveralls. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The method of testing specified in FED-STD-191, wherever applicable and as listed in table VI, shall be followed. Except for the adhesion of gum strapping, cemented seams, and the hydrostatic resistance of cemented seams, all the physical and chemical values specified in section 3 apply to results of each individual determination. The adhesion of gum strapping and cemented seams and the hydrostatic resistance of cemented seams value specified in section 3 apply to lot average of all determinations. The sample unit shall be one of the finished coveralls. The sample size shall be as follows:

Lot size (coveralls)	Sample size (sample units)
280 or less	1
281 to 500	2

NOTE: No lot shall exceed 500 coveralls.

The lot shall be unacceptable if any individual determination or the lot average of all determinations as applicable, fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

TABLE VI. End item tests

Component	Characteristic	Requirement paragraph	Test method	Determination per sample unit	Results reported as
Labels (1 each)	Crocking:	3.3.7	5651 1/		
	Dry			1	Pass or fail
	Wet			1	Pass or fail
Sealed seams	Acid resistance	3.3.12.1	4.5.4.3	2	Pass or fail
	Surface tack after acid exposure	3.3.12.1	4.5.4.5	2	Pass or fail
	Fuel resistance	3.3.12.2	4.5.4.4	2	Pass or fail
	Surface tack after fuel exposure	3.3.12.2	4.5.4.5	2	Pass or fail
Gum strapping	Adhesion	3.3.13	4.5.1	3	Numerically to nearest 0.1 lb./in.
	Adhesion after immersion	3.3.13	4.5.1.1	3	Numerically to nearest 0.1 lb./in.
	Adhesion after heat aging	3.3.13	4.5.1.2	3	Numerically to nearest 0.1 lb./in.
Cemented seams	Adhesion	3.3.13	4.5.2	3	Numerically to nearest 0.1 lb./in.
	Adhesion after immersion	3.3.13	4.5.2.1	3	Numerically to nearest 0.1 lb./in.
	Adhesion after heat aging	3.3.13	4.5.2.2	3	Numerically to nearest 0.1 lb./in.
	Resistance to cold crack	3.3.13	4.5.3	1	Pass or fail
	Hydrostatic resistance	3.2.13	5512 2/	5	Numerically to nearest 1 p.s.i.

1/ The label shall be cut from the coveralls to give a test specimen 4 by 8 inches with the legend centrally located and with the printing perpendicular to the lengthwise direction, and so positioned within the specimen as to be encompassed in the test area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

2/ Hydrostatic pressure shall be applied to the outside of the seam with seam centered directly over the opening.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of unit packs is more or less than required. Number of coveralls per unit pack is more or less than required. 1/ Size shown in one or more coveralls not as specified on unit pack or outer container. 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection.

4.5.1 Seam gum strapping adhesion test. The adhesion of the gum strapping to the seam shall be determined as specified in Method 5962 of FED-STD-191, except as follows: The test sample shall age at least 48 hours after application of the strapping and vulcanization of the assembly before testing. Only straight seamed and strapping section shall be tested. The tested specimen shall have the gum strapping peeled back from the seam for a distance of 2-1/2 to 3 inches. The end of the separated gum strapping shall be clamped in the upper jaw and the lapped seam, with strapping removed, clamped in the lower jaw. The results shall be calculated to the nearest 0.1 pound per inch of width.

4.5.1.1 Seam gum strapping adhesion after immersion test. The strapping shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at 80° ± 9°F, removed from the water, and while still dripping wet, tested for seam strapping adhesion as specified in 4.5.1.

4.5.1.2 Seam gum strapping adhesion after heat aging test. Samples shall be exposed for 7 days in a circulating air oven at 158° ± 2°F, removed from the oven, conditioned for 24 hours at standard conditions and then tested for seam strapping adhesion as specified in 4.5.1.

4.5.2 Cemented seam adhesion test. The samples used in the performance of the test specified in 4.5.1 (with gum strapping removed) shall be used as the test samples for the test. The adhesion of the cemented seam shall be determined as specified in Method 5960 of FED-STD-191. The results shall be calculated to the nearest 0.1 pound per inch of width.

4.5.2.1 Cemented seam adhesion after immersion test. The seam shall be measured for 2 hours in boiling water, removed, immersed for 15 minutes in water at 80° ± 9°F, removed from the water, and while still dripping wet, tested for seam adhesion as specified in 4.5.2.

4.5.2.2 Cemented seam adhesion after heat aging test. Samples shall be exposed for 7 days in a circulating air oven at 158° ± 2°F, removed from the oven, conditioned for 24 hours at standard conditions, and then tested for seam adhesion as specified in 4.5.2.

4.5.3 Resistance of finished seams (cemented and strapped) to cold crack test. The test shall be performed as specified in Method 5874 of FED-STD-191 with the following exceptions: Test temperature shall be -40° ± 2°F. Test specimens shall be conditioned for not less than 4 hours at test temperature prior to testing. Test specimens shall be prepared by folding so that two seam surfaces are superimposed. The roller weight shall be passed along the fold and across the seam. The test specimen shall be visually examined for failure after test. No hydrostatic test shall be performed after cold test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

4.5.4 Chemical resistance tests of finished sealed seams.

4.5.4.1 Apparatus and chemicals.

(a) Test fixture No. 2 (see Drawing 8-1-3).
(b) Wax - shall be paraffin wax or a mixture of paraffin and beeswax containing not more than one-third beeswax.
(c) Petroleum jelly.
(d) Aluminum block - dimensions approximately 4 by 2-7/16 by 1/2 inches.
(e) Chemical agents detector paper - M9 conforming to MIL-P-51493.
(f) Propellant nitric acid types IIIA or IIIB conforming to MIL-P-7254 (see 6.5).
(g) Glass supporting and cover plates measuring 8 by 8 inches.
(h) Distilled water.
(i) Onion skin paper conforming to type V of UU-P-328.
(j) Roller - approximately 5-1/2 inches in diameter, 2 inches wide, and 10 pounds in weight.
(k) Calcium chloride, anhydrous, conforming to type II, grade A of A-A-169.

4.5.4.2 Preparation of test specimens. The test specimen shall be a section, 6 inches in diameter, including a length of finished strapped seam equal to the specimen diameter. The test specimen shall be placed in the test fixture (see Drawing 8-1-3) in the following manner:

(a) Place the test specimen between the spacer inserts and seal with melted wax. The wax shall form a continuous bead covering the outer circumference of the insert assembly.

(b) Place the insert assembly in the test jig sleeve and fill the cavity between the insert assembly and the sleeve wall with melted wax. Mate the jig sleeve and jig collar and screw hand tight.

(c) Seal the inside of the sleeve collar assembly with melted wax: The wax seal shall be continuous and shall extend around and out onto the test specimen approximately 1/8 inch, and up to the inside of test jig to a point approximately 1/8 inch above the top of the spacer insert. Turn the jig assembly over and seal the bottom cavity with melted wax in similar fashion.

(d) During chemical exposure tests, the top of the test apparatus shall be covered with a glass plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

4.5.4.3 Acid resistance test of the finished sealed seam and surrounding area.

(a) Prepare test specimen as specified in 4.5.4.2.

(b) Place test assembly in desiccator over calcium chloride, anhydrous conforming to type II, grade A of A-A-169 and condition at 70° ± 2°F for a minimum of 16 hours.

(c) After conditioning, remove from desiccator and immediately perform acid exposure test as follows:

(1) Coat bottom and top edge of the test fixture with 1/32 inch of petroleum jelly.

(2) Place aluminum block in the center of a glass supporting plate with a piece of the vesicant detector paper, coated side up, on top of the block. Place the test assembly over the block with the heavily coated side of the test specimen up.

(3) Cover the test specimen with 50 milliliters of propellent nitric acid and cover the top of the test fixture with a glass cover plate.

(d) After 2 hours acid exposure at 70° ± 2°F and 65 percent ± 2 percent relative humidity, remove the test specimen from the test fixture, rinse in distilled water, dry at stated conditions for approximately 4 hours and examine for the requirements in 3.3.12.1.

4.5.4.4 Fuel resistance test of the finished sealed seam. The test specimen, prepared as specified in 4.5.4.2 and conditioned as specified in 4.5.4.3 shall be covered with 50 milliliters of JP-4 of MIL-T-5624 for 15 minutes at conditions of 70° ± 2°F and 65 percent ± 2 percent relative humidity. After fuel exposure, the test specimens shall be removed from the test fixture, rinsed in distilled water, dried for approximately 4 hours at the stated conditions and examined for conformance to the requirements in 3.3.12.2.

4.5.4.5 Surface tack rating of the finished sealed seam after acid or fuel exposure. The surface tack test shall be conducted, under conditions of 70° ± 2°F and 65 percent ± 2 percent relative humidity as follows:

(a) The test specimen, prepared as specified in 4.5.4.2 and conditioned as specified in 4.5.4.3 shall be covered with 50 milliliters of the specified acid or fuel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

(b) After 15 minutes exposure, under conditions of 70° ± 2°F and 65 ± 2 percent relative humidity, the specimen shall be removed from the test fixture, rinsed in distilled water, and dried for approximately 4 hours at stated conditions.

(c) Surface tack rating shall be determined as follows:

(1) The specimen, with wax line removed by cutting or scraping, shall be placed exposed side up on a smooth hard surface, and covered with a 2 by 4 inch piece of onionskin paper.

(2) The roller shall be passed along the full length of the onionskin paper covered seam twice in each direction (four passes) with pressure applied by the weight of the roller only.

(3) After rolling, the onionskin paper shall be peeled from the test specimen and the surface tack rating determined by comparison with figure 2 and rating standards.

Rating standards 1/

Rating I - A rating of I shall be equivalent to no trace 2/ of coating material retained on the test paper or no trace of paper fiber adhering to test specimen surface.

Rating II - A rating of II shall be equivalent to a trace 2/ of coating material retained on the test paper or a trace of paper fiber adhering to the test specimen surface.

Rating III - A rating of III shall be equivalent to any degree of coating material or paper fiber transfer in excess of rating II.

1/ Figure 2 pictorially represents the Rating Standards.

2/ A trace is defined as coating material or paper fiber transfer covering 25 percent of test specimen area.

5.. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Two pairs of coveralls of one size only, folded and positioned as shown on Drawing 2-1-293 shall be completely wrapped in 30-pound minimum basis weight kraft paper conforming to A-A-203. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

coveralls shall be completely separated by a sheet of the specified kraft paper prior to wrapping. The wrapped coveralls shall be unit packed in a fiberboard box conforming to style RSC, type CF, (variety SW), class domestic, grade 175 of PPP-B-636. Inside dimensions of the box shall be 24 inches in length, 15-1/2 inches in width, and 5 inches in depth. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial preservation. Coveralls shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Four pairs of coveralls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 25 inches in length, 16-1/2 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for the type II, class F loads.

5.2.2 Level B packing. Four pairs of coveralls of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level B unit packs shall be packed flat, two in depth within the shipping container. Inside dimensions of each shipping container shall approximate 25 inches in length, 16-1/2 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Coveralls, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

5.3 Palletization. When specified (see 6.2), coveralls of one size only, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coveralls are intended for wear by military personnel in contact with guided missile propellants as a full protective outer covering.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Size required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.1, 4.3, and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Type and class of unit load (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples address the contracting activity issuing the invitation for bids or request for proposal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

6.5 Conductance of acid resistance tests. Because of the danger involved in handling and shipping of propellant nitric acid, the acid resistance test specified in 4.5.4.3 will be performed by the government. The samples to be tested for acid resistance should be forwarded to the following address:

Commander
Defense Personnel Supply Center
Clothing and Textile Laboratory Branch
ATTN: DPSC-TTL
2800 South 20th Street
Philadelphia, PA 19101

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8415-0732)

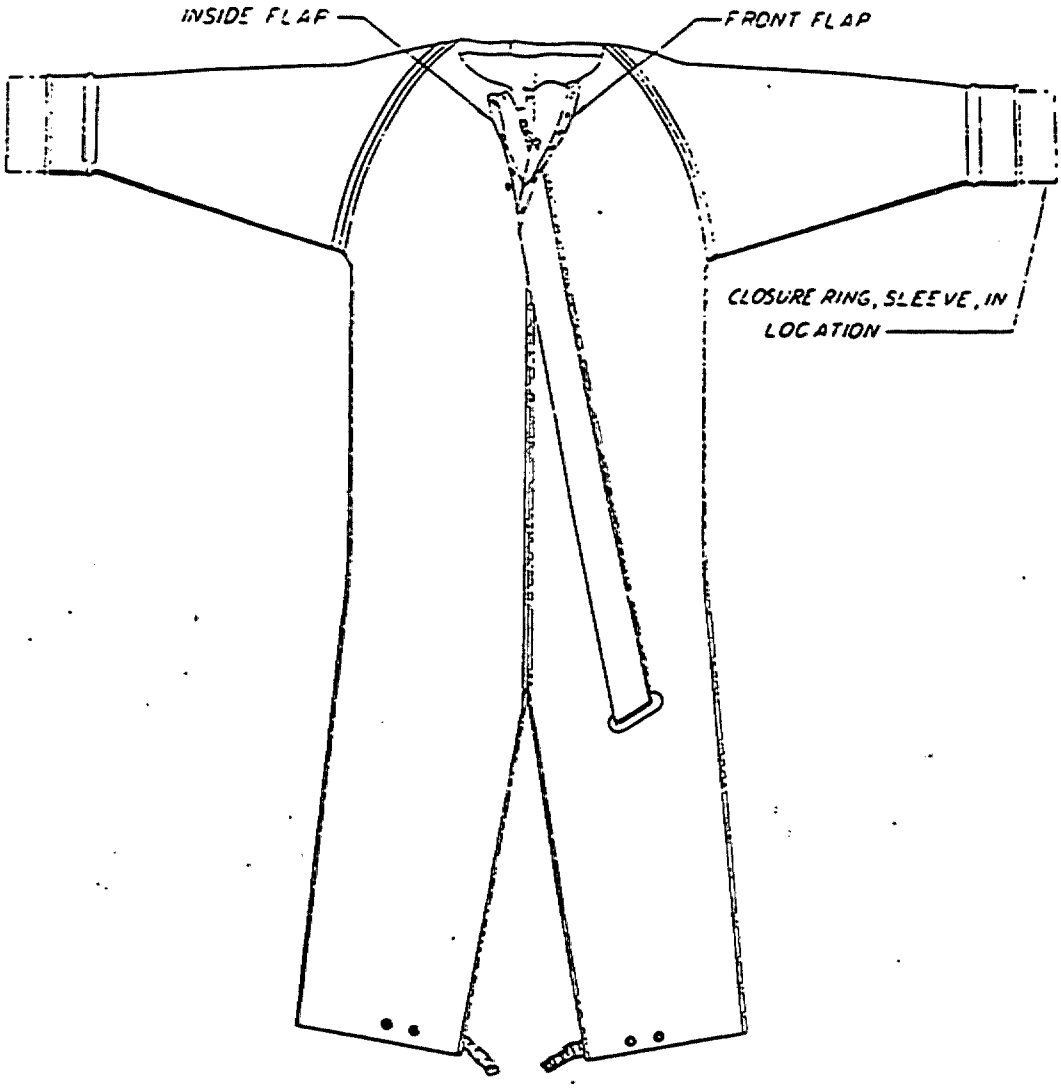
Review activities:

Army - MD
Air Force - 82
DLA - CT

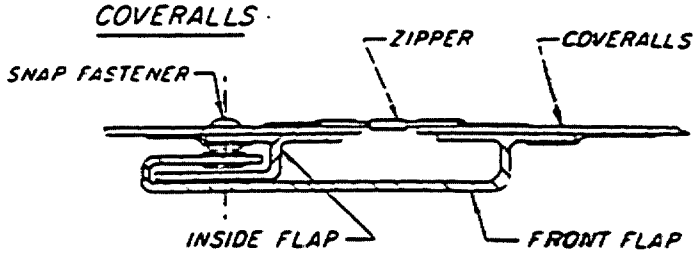
User activity:

Air Force - 45

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.



COVERALLS



SECTIONAL VIEW THRU CLOSURE OF COVERALLS

FIG 1 - COVERALLS, ROCKET FUEL HANDLERS, IMPERMEABLE, FULL PROTECTION

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.

RATINGS

I II III

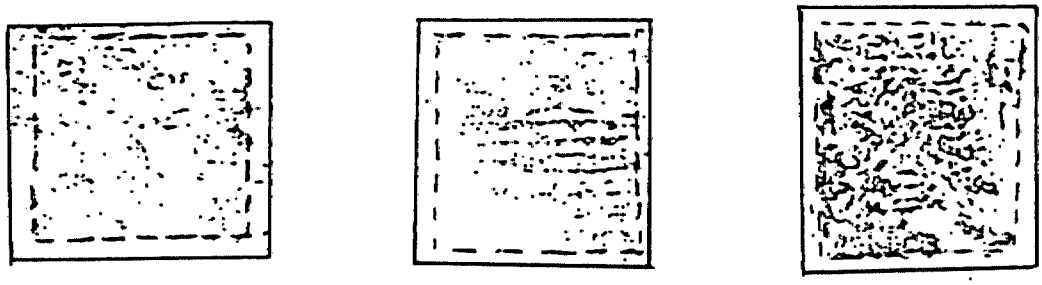


FIGURE 2 SURFACE TACK STANDARDS

NITRIC ACID TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:29Z
Check the source to verify that this is the current version before use.